COMP9517: Computer Vision

2026 T3 Lab 3 Specification

Maximum Marks Achievable: 2.5

This lab is **worth 2.5% of the total course marks**.

The lab files should be submitted online. Instructions for submission will be posted closer to the deadline. **Deadline for submission is Week 5, Friday 16 October 2026, 18:00:00 AET.**

Objective: This lab revisits important concepts covered in the lectures of Week 4 and aims to make you familiar with implementing specific algorithms.

Software: You are required to use OpenCV 3+ with Python 3+ and submit your code as a Jupyter notebook (see coding and submission requirements below). For this lab you will also need to use SciKit-Learn (see links below). In the tutor consultation session this week, you can ask any questions you may have about this lab.

Materials: The dataset to be used in this lab is BloodMNIST, which is part of the bigger MedMNIST dataset: https://medmnist.com/. It consists of microscope images of 8 different types of blood cells, with in total 17,092 images and corresponding labels (11,959 for training, 1,712 for validation, and 3,421 for testing). Each image is 28 x 28 pixels. The dataset is available on WebCMS3 or via the original website (linked above).

Submission: All code and requested results are assessable after the lab. Submit your source code as a Jupyter notebook (.ipynb file) that includes all output and answers to all questions (see coding requirements at the end of this document) by the above deadline. The submission link will be announced in due time.

Task (2.5 marks)

The goal of this lab is to implement and compare the performance of a K-Nearest Neighbours (KNN) classifier, a Decision Tree (DT) classifier, and a Stochastic Gradient Descent (SGD) classifier on the given dataset.

Step 1: Import relevant packages

We will mainly be using SciKit-Learn for this lab, so make sure you have correctly installed this library before moving to the next steps. Check the following link to learn more about the

library and ways to install it: https://scikit-learn.org/stable/

Check the following links on how to import KNN, DT, and SGD classifiers:
https://scikit-learn.org/stable/modules/generated/sklearn.neighbors.KNeighborsClassifier.html
https://scikit-learn.org/stable/modules/generated/sklearn.tree.DecisionTreeClassifier.html
https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.SGDClassifier.html

Step 2: Download the dataset

Download the dataset from WebCMS3 or from the original website (linked above) and familiarize yourself with it. Specifically, you can check how many images and labels there are in the entire dataset, what the size of each image is, how many classes there are and how many images are in each class. Also, have a look at some images of each label.

Step 3: Split and sample the dataset

There are 17,092 images in the dataset. To reduce computational cost, we can work with a subset of the complete dataset. Initially, take 5,000 randomly sampled images from the training subset and 1,000 randomly sampled images from the testing subset.

Make sure that random sampling is done in a stratified way. That is, it must be done such that the number of images in each class ends up being approximately the same. This is to avoid class imbalance in training and testing.

Also make sure the testing images are kept strictly separated from the training set. This is to avoid leakage from the training to the testing results (it is easier for a classifier to correctly classify a sample from the training set than a new sample it has never seen before).

After completing all the steps below, increase the number of training images from 5,000 to all 11,959, and the testing images from 1,000 to all 3,421, and repeat the experiment to see if it makes a difference in classification performance. Note that, unlike in the previous experiment where we ensured the dataset was balanced, the classes in the full dataset are not balanced. There is nothing we can do about that.

Step 4: Perform necessary data reshaping

Each image in the dataset is 28 x 28 pixels. To be able to use the images as input for the machine learning classifiers, you need to reshape (flatten) them.

Step 5: Initialise the classifier

For each classifier (KNN, DT, SGD), initialise the classifier object. It is important to read the documentation to learn about various parameters that can be configured when initialising classifiers. For KNN, set k to 3. For SGD, set the maximum number of iterations (epochs) to 250. Use default settings for all other parameters.

Step 6: Fit the classifier to the training data

The SciKit-Learn library has a fitting method to learn from data. Use the `fit()` method to train each classifier by passing the training data and the training labels as parameters.

Step 7: Evaluate the trained model on the testing data

After you have trained a classifier (also called a model), you can use it to make predictions on the testing data. Use the `predict()` method provided by the SciKit-Learn library.

Step 8: Report the performance of each classifier

To quantify how each of the trained classifiers performs, use standard classification metrics such as accuracy, precision, recall, and the F1-score. To summarise the errors for each class, use the confusion matrix.

The SciKit-Learn library provides built-in methods to automatically calculate these measures by comparing the predicted labels with the provided ground-truth labels. Click on the following link to find these methods and import them:
https://scikit-learn.org/stable/modules/model_evaluation.html

For each classifier (KNN, DT, SGD) and training sample size (both the smaller subset and the full set), show the values of all of the abovementioned standard classification metrics and the confusion matrix in your Jupyter notebook.

Coding Requirements

Make sure that in your Jupyter notebook, all outputs and other requested results are displayed in the notebook environment. All cells in your notebook must have been executed so that the tutor/marker does not need to execute the notebook again to see the results.

Copyright: UNSW CSE COMP9517 Team. Reproducing, publishing, posting, distributing, or translating this lab assignment is an infringement of copyright and will be referred to UNSW Student Conduct and Integrity for action.

Released: 6 October 2026